KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

Certification Guide

Tier 1 — Basic Repair

T1

Safe hands, the right tools, and the repairs every Tech Bench student does first.

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Name: _____ Started: _____

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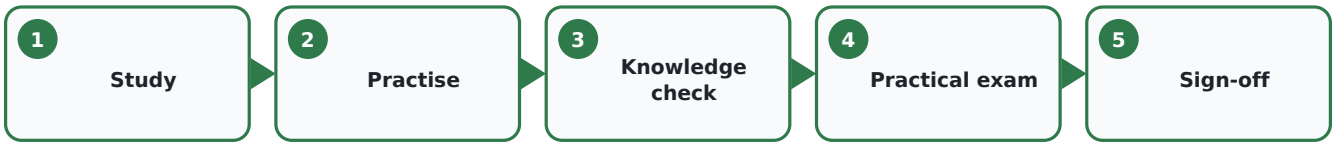
1 About this certification

What it lets you do

- Take Tier 1 tickets on your own: keyboards, trackpads, bezels, batteries, chargers and cases.
- Do the four software fixes (hard reset, diagnostics, powerwash/recovery, keyboard layout) without supervision.
- Open a bottom cover or keyboard deck and reseat connectors.
- Help a Tier 2 or Tier 3 tech by doing the opening and closing steps of bigger repairs.

Before you start. None. This is the first repair certification.

How to earn it



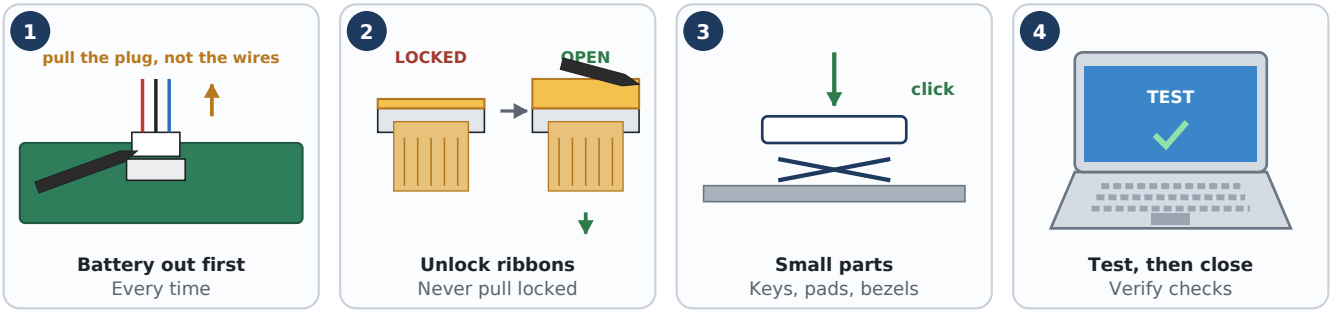
Step	What you do
Study	Read the Basic Skills chapter in any repair guide, and Modules 1–4 below.
Practise	Do 6 supervised Tier 1 repairs (see the practice log). At least one battery, one keyboard, and one software fix; at least two different brands.
Knowledge check	Score 80% or better (12 of 15) on the written check.
Practical exam	Pass the observed practical exam with no automatic-fail items.
Sign-off	Lead tech or manager signs the last page and awards the certification in the portal.

Certification is earned, not given for time served. Every requirement below must be signed off by a lead tech or manager.

A certification can be paused by a manager after a safety violation or repeated comebacks (repairs that come back with the same fault).

When every box is signed, the manager awards the certification on the portal's Certifications page. That's what unlocks the matching tickets and guides.

2 What you need to know



Battery out first
Every time

Unlock ribbons
Never pull locked

Small parts
Keys, pads, bezels

Test, then close
Verify checks

Repairs in this tier, and where the guides are

Open the device in the portal's Learn tab and pick the topic. Each topic opens at the right section of that model's repair guide. Read the device's guide every time: screw counts and steps differ between models.

Learn tab topic	Covers
Keyboard Replacement (quick-swap models)	Keyboards, single keys
Trackpad Replacement / Trackpad Cable Replacement	Touchpad and its cable
Bezel Replacement	Display bezel
Battery Replacement	Battery
Bottom Assembly Replacement	Opening and refitting the bottom cover
Device Powerwash, Diagnostics, Keyboard International Registration	The four software fixes (S1–S4)
(any device page) Reseating internal connectors	Connector reseat

Module 1 — Safety and the bench

- **Power and battery.** Power off, unplug, remove the SD card. Inside, unplug the battery before anything else and plug it back in last. A swollen, hot, hissing or sweet-smelling battery is a stop: don't touch it, move away, get an adult.
- **Static.** Wear the wrist strap clipped to the ESD mat. Touch boards only by their edges.
- **Tools.** Plastic only near electronics (picks, spudgers). Metal tweezers and drivers never pry. Phillips #0 and #1 cover almost every Chromebook screw.
- **Screws.** Lay them out on the mat in the same pattern as the device. Long screws only go back in long holes. Snug, not tight.
- **Photos.** Photograph every layer before you remove it. Your photos are your reassembly instructions.

Module 2 — What counts as Tier 1

- The district fee schedule puts these in **Tier 1 (Basic Repair)**: keyboard, trackpad, bezel, battery, charger / AC adapter, protective case, and other small parts as assessed.
- Two or more Tier 1 items on the same device can move the claim to Tier 2. If a ticket lists more than one part, ask the lead tech what tier it is before starting.
- On some models a keyboard is a quick swap; on others it's built into the palm rest and means moving every part (a Level 3 job in the repair guide). Check the device's guide first. If the guide marks it Level 3, it's not yours yet, even though the fee tier is 1.

Module 3 — Diagnose before you replace

- Start with the cheapest fix: hard reset (S1), then diagnostics (S2), then a connector reseat. Many "broken" Chromebooks need no part at all.
- Wrong symbols when typing is almost always a keyboard *layout* (S4), not a broken keyboard.
- Won't turn on: known-good charger for 30 minutes, try both USB-C ports, then hard reset. A battery that's been unplugged needs the charger to wake up.
- Write what you found and what you tried in the ticket notes before ordering a part.

Module 4 — Finishing a repair properly

- Test with the cover still off, before the last screws go in.
- Run the repair's "Verify before closing the ticket" checks every time.
- Count your screws: none left over.
- Update the ticket: what you did, the part used, the new status. Log the part on the parts sheet.
- Hand the device to a mentor or manager for check-off if the bench requires it.

3 Skills checklist

An evaluator initials each skill after watching you do it correctly, on your own, at least once. Practice repairs count.

#	Skill	Date	Evaluator
1	Sets up the bench: strap on, mat clear, screw tray, photos		
2	Picks the right driver bit and removes screws without stripping		
3	Opens a bottom cover (or keyboard deck) with plastic tools only		
4	Unplugs and replugs the battery correctly (first out, last in)		
5	Unlocks, removes and reseats a ZIF ribbon cable		
6	Replaces a single keycap		
7	Replaces a battery and wakes the device with the charger		
8	Replaces a keyboard (quick-swap model) or trackpad		
9	Replaces a display bezel without cracking it		
10	Runs a hard reset, diagnostics, powerwash and recovery		
11	Fixes a wrong keyboard layout		
12	Writes clear ticket notes and logs the part used		

4 Supervised practice log

Log every supervised practice repair (or front-desk transaction). The supervisor signs each line.

#	Date	Device / asset tag	Repair or task	Ticket #	Supervisor
1					
2					
3					
4					
5					
6					

5 Knowledge check

15 questions. Circle one answer for each. Closed book. The evaluator marks it with the key. Pass mark is on page 1.

1. What is the first thing you unplug once the bottom cover is off?

A The keyboard ribbon
B The Wi-Fi antennas
C The battery
D The speaker cable

2. A student's keyboard types @ where " should be. What do you try first?

A Check the keyboard layout (S4)
B Replace the keyboard
C Reseat the keyboard ribbon
D Powerwash right away

3. Which tool should you use to open a plastic seam?

A Flat-head screwdriver
B Metal tweezers
C Pocket knife
D Plastic pick

4. A battery looks puffy and the case is bulging. What do you do?

A Press it flat and continue
B Stop, don't touch it, get an adult
C Pry it out quickly
D Charge it to check it

5. How do you release a ribbon cable from a ZIF connector?

A Pull the ribbon firmly
B Flip the latch up first, then slide it out
C Cut the ribbon
D Pry the connector off the board

6. You replaced a battery and the Chromebook won't turn on. Most likely fix?

A Order a new board
B Replace the screen
C Reinstall the old battery
D Plug in the charger and wait

7. Where do long screws go back?

A Only in the long-screw holes
B Anywhere they fit
C Near the battery
D They don't go back

8. Which of these is a Tier 1 repair on the district schedule?

A Screen
B Motherboard
C Trackpad
D Hinge

9. A device won't turn on. What do you do before opening it?

A Order a battery
B Replace the motherboard
C Powerwash
D Charger 30 min, try both ports, hard reset

10. When do you test a repair?

A After closing everything
B With the cover still off, before the last screws
C Only if the student complains
D Never; the part is new

11. What does the wrist strap protect against?

A Burns
B Cuts
C Static damage to boards
D Drops

12. You have one screw left over after reassembly. What now?

A Find where it goes before closing the ticket
B Throw it away
C Tape it to the device
D Ignore it

13. Powerwash erases...

A Files saved on the device (like Downloads)
B Nothing
C Google Drive files
D The enrollment

14. The device's guide marks its keyboard replacement as Level 3. You hold Tier 1. What do you do?

A Do it anyway; keyboards are Tier 1
B Skip the hard steps
C Pass it to a Tier 3 tech or lead tech
D Close the ticket

15. What goes in the ticket notes?

A Nothing; the status is enough
B Only the student's name
C Your hours
D What you found, what you tried, the part used

Score	Pass mark	Result	Evaluator / date
____ / 15	80% or better (12 of 15) on the written check	Pass / Retake	

6 Practical exam

An evaluator watches you do each task. Time allowed: about 45 minutes. You may use the repair guides and your notes, but you work on your own.

Task	What you'll do	To pass	P / F
Bench setup and safety	Set up the bench and power down a test device correctly.	Strap on, mat clear, screw tray, photos taken, device off and unplugged.	
Open and battery	Remove the bottom cover (or deck) and unplug the battery.	Plastic tools only, screws laid out in pattern, battery unplugged first, by the plug body.	
Connector drill	Unlock, remove, inspect and reseat two different ribbon cables.	Latch opened before pulling, ribbon in straight and fully, latch closed.	
Tier 1 repair	Do one Tier 1 repair chosen by the evaluator (battery, keyboard, trackpad or bezel).	Follows the device guide, tests before closing, verify checks pass.	
Software fix	Diagnose a planted software fault (wrong layout or a device that needs a hard reset).	Finds and fixes it without opening the device.	
Close-out	Reassemble, account for every screw, and write the ticket notes.	No leftover screws, clear notes, part logged.	

Automatic fail. Any one of these ends the exam: Prying with a metal tool against the battery or a board; Working inside with the battery connected; Pulling a locked ribbon cable; Ignoring a swollen battery; Leaving screws unaccounted for. You can retake it after a week of extra practice.

7 Sign-off

Requirement	Done	Signed
Study	☐	
Practise	☐	
Knowledge check	☐	
Practical exam	☐	
Prerequisite: None. This is the first repair certification.	☐	

Candidate	
Evaluator (lead tech or manager)	
Date awarded	
Awarded in the portal (Certifications page) by	

Managers: award it on the portal's Certifications page so it shows on the student's portfolio and unlocks the matching tickets and guides.